

Instalar Tomcat9 y configurar Apache

```
root@serveraweb1amp:/home/jens# apt install tomcat9 tomcat9-admin
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
tomcat9 ya está en su versión más reciente (9.0.58-1ubuntu0.2).
tomcat9-admin ya está en su versión más reciente (9.0.58-1ubuntu0.2).
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 0 no actualizados.
root@serveraweb1amp:/home/jens# systemctl status tomcat9
● tomcat9.service - Apache Tomcat 9 Web Application Server
   Loaded: loaded (/lib/systemd/system/tomcat9.service; enabled; vendor preset: en>
   Active: active (running) since Sun 2025-11-23 18:24:19 UTC; 1min 9s ago
     Docs: https://tomcat.apache.org/tomcat-9.0-doc/index.html
   Process: 27411 ExecStartPre=/usr/libexec/tomcat9/tomcat-update-policy.sh (code=e>
   Main PID: 27415 (java)
     Tasks: 22 (limit: 4555)
    Memory: 137.0M
       CPU: 6.260s
    CGroup: /system.slice/tomcat9.service
            └─27415 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.f>

nov 23 18:24:23 serveraweb1amp tomcat9[27415]: Al menos un JAR, que se ha explorado >
nov 23 18:24:23 serveraweb1amp tomcat9[27415]: Deployment of deployment descriptor [>
nov 23 18:24:23 serveraweb1amp tomcat9[27415]: Despliegue del descriptor de configur>
nov 23 18:24:23 serveraweb1amp tomcat9[27415]: The path attribute with value [/host->
nov 23 18:24:23 serveraweb1amp tomcat9[27415]: Al menos un JAR, que se ha explorado >
nov 23 18:24:23 serveraweb1amp tomcat9[27415]: Deployment of deployment descriptor [>
nov 23 18:24:23 serveraweb1amp tomcat9[27415]: Desplegando el directorio [/var/lib/t>
nov 23 18:24:24 serveraweb1amp tomcat9[27415]: Al menos un JAR, que se ha explorado >
nov 23 18:24:24 serveraweb1amp tomcat9[27415]: Deployment of web application directo>
nov 23 18:24:24 serveraweb1amp tomcat9[27415]: Server startup in [3195] milliseconds

root@serveraweb1amp:/home/jens#
```

Configurar memoria en JAVA

```
root@serveraweb1amp:/home/jens
GNU nano 6.2 /etc/default/tomcat9 *
# The home directory of the Java development kit (JDK). You need at least
# JDK version 8. If JAVA_HOME is not set, some common directories for
# OpenJDK and the Oracle JDK are tried.
#JAVA_HOME=/usr/lib/jvm/java-8-openjdk

# You may pass JVM startup parameters to Java here. If you run Tomcat with
# Java 8 instead of 9 or newer, add "-XX:+UseG1GC" to select a suitable GC.
# If unset, the default options will be: -Djava.awt.headless=true
JAVA_OPTS="-Djava.awt.headless=true -Xmx1024m"

# To enable remote debugging uncomment the following line.
# You will then be able to use a Java debugger on port 8000.
#JAVA_OPTS="${JAVA_OPTS} -agentlib:jdwp=transport=dt_socket,address=8000,server=y,suspend=n"

# Java compiler to use for translating JavaServer Pages (JSPs). You can use all
# compilers that are accepted by Ant's build.compiler property.
#JSP_COMPILER=javac

# Enable the Java security manager? (true/false, default: false)
#SECURITY_MANAGER=true

# Whether to compress logfiles older than today's
#LOGFILE_COMPRESS=1
```

Configurar usuarios para Tomcat Manager

```
root@serveraweb lamp: /home/jens
GNU nano 6.2 /etc/tomcat9/tomcat-users.xml *
<?xml version="1.0" encoding="UTF-8"?>
<!--
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
  version="1.0">
  <role rolename="manager-gui"/>
  <user username="admin" password="secret" roles="manager-gui"/>
</tomcat-users>
<!--
```

Conectar Apache con Tomcat (Reverse Proxy)

```
root@serveraweb lamp: /home/jens
root@serveraweb lamp:/home/jens# a2enmod proxy
Enabling module proxy.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@serveraweb lamp:/home/jens# a2enmod proxy_http
Considering dependency proxy for proxy_http:
Module proxy already enabled
Enabling module proxy_http.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@serveraweb lamp:/home/jens# a2enmod rewrite
Module rewrite already enabled
root@serveraweb lamp:/home/jens# a2enmod headers
Enabling module headers.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@serveraweb lamp:/home/jens# systemctl restart apache2
root@serveraweb lamp:/home/jens#
```

Crear una configuración de Apache que redirija a Tomcat

```
root@serveraweb1amp: /home/jens
GNU nano 6.2 /etc/apache2/sites-available/tomcat-proxy.conf *
<VirtualHost *:80>

    # Permite mantener el host original
    ProxyPreserveHost On

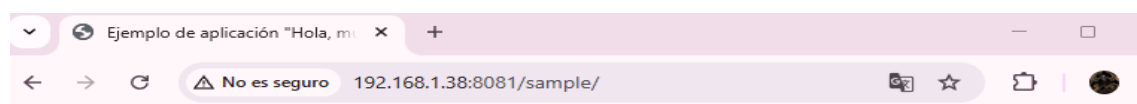
    # Proxy hacia Tomcat
    ProxyPass /app http://localhost:8080/app
    ProxyPassReverse /app http://localhost:8080/app

</VirtualHost>
```

Activar el sitio de Tomcat en Apache

```
root@serveraweb1amp: /home/jens
root@serveraweb1amp:/home/jens# nano /etc/apache2/sites-available/tomcat-proxy.conf
root@serveraweb1amp:/home/jens# systemctl restart apache2
root@serveraweb1amp:/home/jens# a2ensite tomcat-proxy
Enabling site tomcat-proxy.
To activate the new configuration, you need to run:
    systemctl reload apache2
root@serveraweb1amp:/home/jens# systemctl reload apache2
root@serveraweb1amp:/home/jens#
```

Acceder a tomcat en la URL



Ejemplo de aplicación "Hola, mundo"

Esta es la página de inicio de una aplicación de muestra utilizada para ilustrar la organización del directorio de origen de una aplicación web utilizando los principios descritos en la Guía del desarrollador de aplicaciones.

Para comprobar que funcionan, puedes ejecutar cualquiera de los siguientes enlaces:

- A una [página JSP](#) .
- A un [servlet](#) .



Página JSP de aplicación de ejemplo

¡Hola!

Esta es la salida de una página JSP que es parte de la aplicación Hello, World.

-He tenido que cambiarle el puerto al 81, porque el 80 lo usa Apache

```
root@serveraweblamp: /home/jens
GNU nano 6.2 /etc/tomcat9/server.xml
and responses are returned. Documentation at :
Java HTTP Connector: /docs/config/http.html
Java AJP Connector: /docs/config/ajp.html
APR (HTTP/AJP) Connector: /docs/apr.html
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="8081" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443" />
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
           port="8080" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443" />
```

